

QSyncFold: quantum neural network for multidimensional sync-discovery in protein folding

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Abstract

Quantum computing provides alternative encoding and sampling paradigms for protein structure prediction (PSP), but existing quantum-PSP methods are often limited by resource-scaling issues and by discrete or inefficient encodings for continuous coordinates. To address these limitations, we propose QSyncFold, a hybrid quantum-classical neural network framework that combines quantum superposition with differentiable learning. QSyncFold employs ProtaQode to simultaneously achieve reversible continuous-space encoding of residue coordinates and parameterized interaction modeling. This is realized by encoding residue-pair interactions in superposition via a decomposable Any-State RY (ASRY) operator that is efficient for a limited qubit budget. Algorithmically, QSyncFold trades register size for iteration count, reducing the qubit requirement for each iteration from $O(N)$ to $3 + \lceil \log_2 N \rceil$, where N is the number of residues. This design ensures the framework is experimentally viable under NISQ constraints. On short peptide structure prediction, QSyncFold achieved a 5.25-fold improvement in the IDDT metric compared with the Variational Quantum Eigensolver baseline and demonstrated a clear trade-off between qubit budget and convergence speed. While using quantum baselines as the primary comparison, the method performance approaches AlphaFold2 in the short peptide domain, with classical methods serving as background reference. This study advances the precision and methodology of quantum computing in PSP, illustrating a viable pathway for quantum algorithms in biomolecular modeling.

Keywords quantum machine learning, quantum neural network, quantum learning algorithms, protein structure prediction

Introduction

Protein structure prediction (PSP) is the task of inferring 3D conformation from its amino acid sequence [1, 2]. As illustrated in Fig. 1, this process culminates in the polypeptide chain folding to form the tertiary structure, which may further assemble into a quaternary structure, making it an ideal challenge for deep learning methodologies. It remains a central problem in computational biology due to structure largely dictates function, which in turn enables inference of function [3–6] and communications [7–9]. Classical machine-learning methods, such as advanced deep learning models like AlphaFold [10–12], RoseTTAFold [13], and ESMFold [14], have achieved remarkable practical success and provide a natural benchmark for applied PSP research. Nevertheless, the underlying combinatorial explosion of residue-residue interaction patterns and the high-dimensional

continuous search over atomic coordinates pose fundamental challenges, as the scale of the conformational space grows exponentially with sequence length and degrees of freedom. It has spurred the development of new representation and prediction paradigms beyond classical deep learning systems.

Quantum machine learning (QML) is one such novel paradigm [15, 16]. By exploiting quantum superposition and entanglement, QML can represent and manipulate exponentially large amplitude vectors with few qubits, offering a fundamentally different mechanism for addressing data explosion and high-dimensional search problems at a theoretical level. This capability has been widely applied in fields such as computer vision [17–20], graph theory [21, 22], natural language processing [23, 24] and software engineering [25]. Early efforts to apply QML to PSP have taken several approaches,

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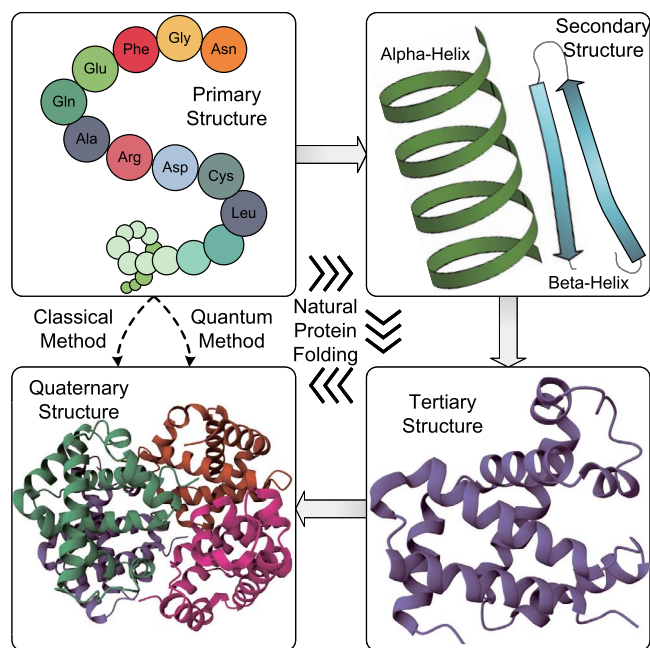


Figure 1. Protein folding process and prediction pathways.

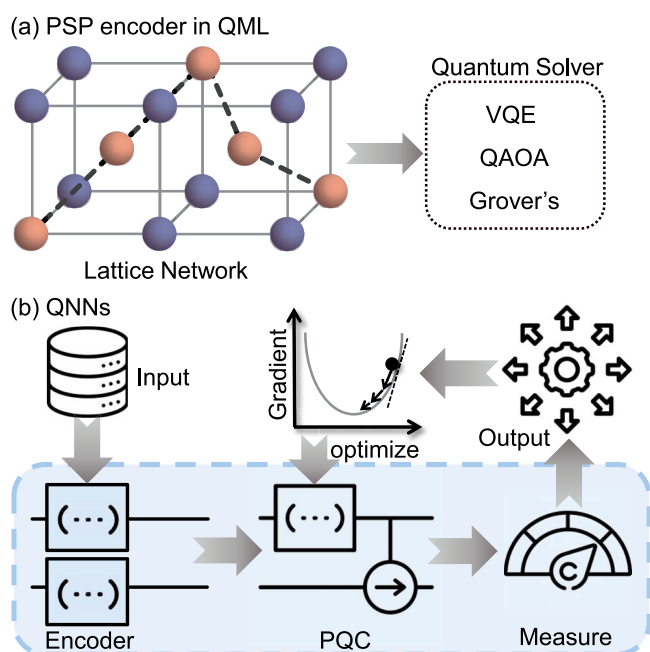


Figure 2. Previous QML encoding and QNN learning methods.

including Hamiltonian-based formulations using Variational Quantum Eigensolvers (VQE) to minimize energy-like objectives [26], Grover-inspired search schemes adapted to folding constraints [27, 28], and hybridizations of classical deep models with quantum walks or other quantum subroutines [29]. Figure 2 shows the lattice network encoding and the QNN workflow employed by existing QML approaches. Although these works confirm the feasibility of quantum approaches on toy or medium-sized instances, they simultaneously expose recurring practical limitations regarding scalable PSP on near-term devices.

Concretely, we identify three interrelated limitations of existing PSP quantum computing methods that hinder their extensibility to realistic modeling tasks. First, resource scaling remains a challenge, as many formulations require a number of qubits that grows linearly with sequence length, such as VQE needing at least $2(n-3)$ qubits and Grover algorithm requiring $3n$ qubits for n residues. This is incompatible with current NISQ hardware and prevents the exploration of moderate-length proteins. Second, lattice-based models failed in representing amino acid spatial arrangements in continuous space, deviating from natural conformations. Third, residue interactions are confined to nearest and next-nearest neighbors in VQE, neglecting long-range effects, while static Hamiltonian parameters hinder adaptability across diverse protein structures. These constraints jointly limit both the fidelity of encoded structures and the practical applicability of PSP quantum methods beyond simple proof-of-concept instances.

Quantum neural networks (QNNs), which are hybrid architectures that parameterize quantum circuits and optimize parameters with classical optimizers, offer a promising route to mitigate these limitations [30, 31]. QNNs are differentiable, can encode continuous variables via parameterized rotations, and permit architectural design that trades qubit budget for quantum circuit and iterative computation.

In this paper, we introduce Quantum Neural Network Multidimensional Sync-Discovery in Protein Folding (QSyncFold), a QNN-based model featuring the Protein Quantum Encoder (ProtaQode) with sub-components Quantum Interactive Residue-Pair Encoder (QIRE) and Quantum Continuous Space Encoder (QCSE). QIRE employs a novel Any-State RY (ASRY) gate to represent residue-pair interactions within superposition systems, while QCSE enables continuous space representation, departing from lattice constraints. The overall workflow is depicted in the Fig. 3. Our contributions are as follows:

- A protein quantum encoding method ProtaQode is proposed, which resolves two critical issues. It uses QCSE to encode continuous coordinates, thereby eliminating lattice discretization, and leverages QIRE to adaptively encode residue-pair interactions in superposition, achieved through a ASRY gate.
- A PSP prediction framework, QSyncFold, is proposed. It employs the ProtaQode encoder and utilizes iterative inference to successfully reduce the qubit resource demand per iteration from $O(n)$ to $3 + \lceil \log_2 N \rceil$.
- Evaluation results demonstrate that QSyncFold consistently outperforms existing PSP quantum models across all metrics. Specifically, on short peptide structure prediction, QSyncFold achieved a 5.25-fold improvement in the lDDT metric compared with the VQE baseline and revealed a clear trade-off between qubit budget and convergence speed.

Material and methods

Decomposition of givens rotation gate

A novel RY gate that can act on any state in superposition is proposed, called any state RY (ASRY), same as Givens Rotation. It aims to influence the amplitudes of two superposition states, similar to the single-qubit operation RY that operates on two states but in a single qubit. Assume all n -qubits are in the superposition state $|\psi\rangle = \sum_{k=0}^{2^n-1} \alpha_k |k\rangle$, and apply the RY gate to the states $|i\rangle$ and $|j\rangle$. An unitary matrix of ASRY gate

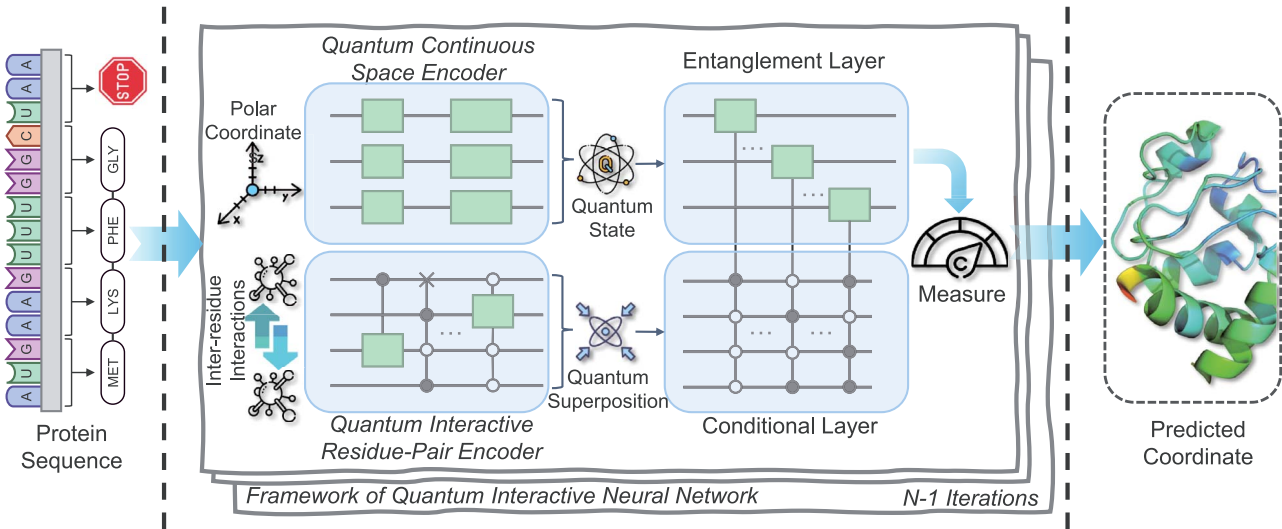


Figure 3. Overview of QSyncFold architecture with quantum encoders and learning circuits.

$U_{ASRY}^{ij}(\theta)$ is delineated in Equation 1, where $U_{ASRY}^{ij}(\theta) \in \mathbb{C}^{2^n}$ is defined.

$$U_{ASRY}^{ij}(\theta) = \begin{bmatrix} 1 & \dots & 0 & 0 & 0 & 0 & 0 \\ \vdots & \ddots & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \cos(\theta/2) & \dots & -\sin(\theta/2) & 0 & 0 \\ & & & 1 & & & \\ 0 & 0 & \vdots & \ddots & \vdots & 0 & 0 \\ & & & & 1 & & \\ 0 & 0 & \sin(\theta/2) & \dots & \cos(\theta/2) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \ddots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \dots & 1 \end{bmatrix}, i < j, \quad (1)$$

where $U_{ASRY}^{ij}(\theta)$ is derived from the identity matrix, and the positions (i, i) and (j, j) are $\cos(\theta/2)$ functions, and the positions (i, j) and (j, i) are $\sin(\theta/2)$ and $-\sin(\theta/2)$ functions. To facilitate a more comprehensive study, regard $U_{ASRY}^{ij}(\theta)$ as RY gate applied state $|i\rangle$ and state $|j\rangle$. ASRY gate will solely alter the amplitude of the corresponding states.

Via quantum gate decomposition Algorithm, ASRY gate will be decomposed to basic quantum gates and control circuits. The basic gates list that the ASRY decomposes is illustrated in Fig. 4. The figure illustrates the four stages required to decompose the ASRY gate by the example of a 4-qubit system, transitioning from the source state $|1111\rangle$ to the target state $|0000\rangle$. Additionally, the pseudocode for the decomposition of the ASRY gate is given in the Algorithm 1, outlining the necessary steps for this process. The ASRY gate enables the application of RY between any two states within a superposition state system, hence enhancing the utilization capabilities of the superposition system. Due to space constraints, the specific definition of the ASRY gate is provided in Supplementary B, its detailed proof of unitarity is in Supplementary C, and the proof of its effectiveness is located in Supplementary D.

Algorithm 1 The decomposition algorithm of ASRY.

Require: the number of qubits N , the source state $|S\rangle$, the target state $|T\rangle$, the parameter θ .

Ensure: a list U of decomposed operations by custom ASRY gate.

- 1:
- 2: **Step 1:** Rearrange the order of source and target state.
- 3: If $S > T$:
- 4: $\theta \leftarrow -\theta$
- 5: $S, T \leftarrow T, S$
- 6: **Step 2:** Determine whether there is only one qubit.
- 7: If $N = 1$:
- 8: $U = \{U, RY(\theta)\}$
- 9: return U
- 10: **Step 3:** Compute the binary difference between S and T .
- 11: $D \leftarrow \{D, i\}$ For i in N If $S[i] \neq T[i]$
- 12: **Step 4:** Flip the non-first-order difference qubit.
- 13: For i in $D[1:]$:
- 14: $U = \{U, X(i).ctrl(\text{all qubits except } i, S \text{ except } i - th)\}$
- 15: $S[i] \leftarrow S[i]$
- 16: **Step 5:** Guarantee the order of the first different qubit.
- 17: If $S[D[0]] = 1$:
- 18: $U = \{U, X(D[0]).ctrl(\text{all qubits except } D[0], S \text{ except } D[0] - th)\}$
- 19: $D[0] \leftarrow th$
- 20: **Step 6:** Make RY on the first different qubit and take adjoint matrix to re-swap.
- 21: $R = RY(D[0]).ctrl(\text{all qubits except } D[0], S \text{ except } D[0] - th)\}$
- 22: $U = \{U, R, reverse(U)\}$
- return U

Protein quantum encoder ProtaQode

ProtaQode is an encoder designed to map the information of protein into quantum states, facilitating their use as inputs for QNNs. This encoder encompasses the coding of positional information as well as the encoding of interaction intensities. Quantum interactive residue-pair encoder, QIRE, is an innovative technique employing

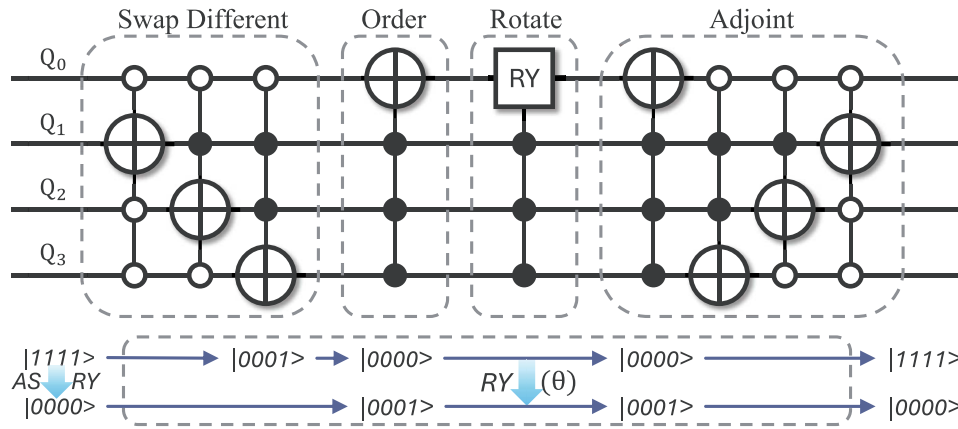


Figure 4. ASRY gate decomposition for state. The example realizes RY in the 4-qubit superposition system from state $|1111\rangle$ to state $|0000\rangle$.

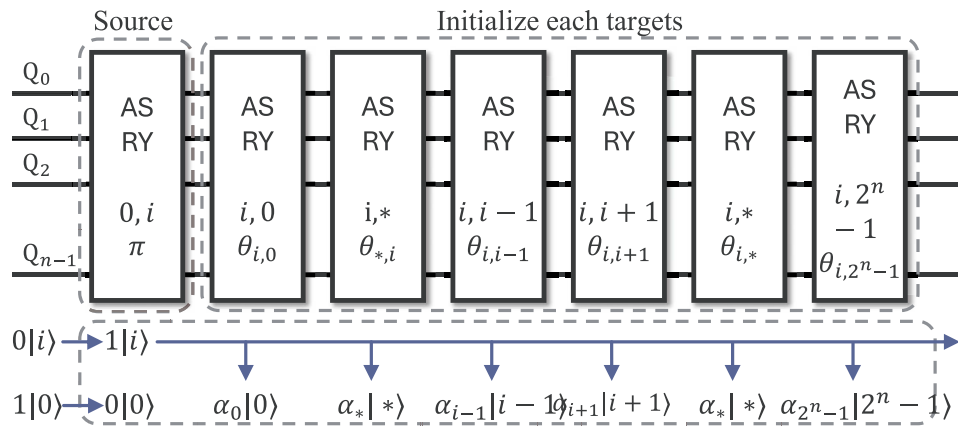


Figure 5. QIRE block quantum circuit, which establishes the interaction strength among residues, illustrating how the amplitude of state $|i\rangle$ is diminished while the amplitude of an interacting state is concurrently augmented.

superposition states to denote the interactive strength between residues. Protein is a biological macromolecule composed of amino acids as its basic building blocks. The interactions among various residues determine the 3D structure and functionality of proteins [32]. However, the interactions of natural states change based on environmental conditions, residue types, sequence positions, and the impact of adjacent residues. Consequently, employing learnable parameters to denote interaction intensities can encapsulate the dynamic behavior of proteins across varying environments. The adaptability is not provided by conventional static models. At the same time, the superposition states can significantly reduce qubits and contribute to the subsequent prediction of structures. Proteins involve 20 unique amino acids. An unidentified amino acid X is incorporated as a supplement to accommodate atypical circumstances. The one-letter symbols for all amino acids constitute the set $A = \{X, C, M, F, I, L, V, W, Y, A, G, T, S, N, Q, D, E, H, R, K, P\}$. Consider a single-chain protein $P = \{r_0, \dots, r_{N-1} \mid r_i \in A\}$ with N residues. For a given residue r_k in protein P , obtain the interaction strength set $I = \{f(< r_k, r_i >) \mid 0 \leq i < N \wedge i \neq k\}$, where the function f calculates the interaction. The Fig. 5 illustrates the quantum circuit of QIRE. For a model with a maximum support length N of a protein, the residue r_k shares interaction relationships with another residue of $N - 1$. Encoding the interactions by quantum superposition states requires $N_q = \lceil \log_2 N \rceil$ qubits, with $2^{N_q} - 1$ states

allocated for interactions and additional k th state for distributing the interaction parameters. Assume the initial system state $|\psi\rangle = |0\rangle$. Subsequently, apply $U_{ASRY}^{0,k}(\pi)$ to the system, which swaps the amplitude and results in the system state of $|\psi\rangle = |k\rangle$. Subsequently, beginning with the source state $|k\rangle$, $U_{ASRY}^{k,i}(\theta_{k,i})$ is employed to distribute the interaction parameters to each other target $|i\rangle$, where $0 \leq i < N$ and $i \neq k$, with $\theta_{k,i}$ being a learnable parameter. Depending on the interaction intensity, the amplitude of $|k\rangle$ is transferred to additional target states $|i\rangle$, representing the quantum encoding of the interaction strength from r_k to r_i . Next, we will examine the correlation between the ASRY parameters and the interaction strength. To minimize the number of parameters, let $\zeta = \{\zeta_i \mid i \in A\} = \{\zeta_X, \dots, \zeta_P\} \in [-1, 1]$ represent the parameter of the residues self-property. Then, compute the initial parameter of the quantum gate using multiplication. Subsequently, apply min-max normalization for scaling. For the residue pair $< r_k, r_i >$, the ASRY parameter $\theta_{k,i}$ is expressed in Equation 2.

$$\theta_{k,i} \leftarrow \frac{\zeta_{r_k} \zeta_{r_i} \sigma}{d_{k,i}} \rho_{|k-i|} \quad (2)$$

Here, $d_{k,i}$ denotes the distance between r_k and r_i , σ signifies a scale parameter, and the parameter $\rho_{|k-i|}$ indicates the scaling of different hop times from i to k . At this point, the interaction strength is initialized

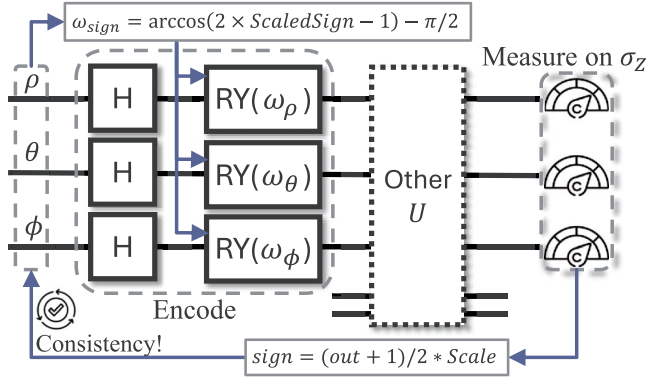


Figure 6. QCSE block quantum circuit, which encodes continuous polar coordinates into quantum states and is fully reversible between input and output operations.

using learnable parameters, as described in the formula Equation 3.

$$\begin{cases} |\psi\rangle \leftarrow U_{ASRY}^{0,k}(\pi) |0\rangle = |k\rangle \\ |\psi\rangle \leftarrow \prod_{i=0, i \neq k}^N U_{ASRY}^{k,i}(\theta_{k,i}) |\psi\rangle = \sum_{i=0}^{2^N-1} \alpha_i |i\rangle \end{cases} \quad (3)$$

The larger the α , the more positive the interaction is in the protein structure, whereas the smaller the α , the more negative the interaction turns out.

The encoder approach of QIRE facilitates adaptive and resource-efficient interaction encoding, which will enhance the model comprehension of the interactions among residues and their positional relationships.

The quantum continuous space encoder (QCSE) is an innovative approach for encoding the relative position of residues into qubits, allowing the representation of continuous space rather than discrete space, such as lattice networks. In QCSE, the relative coordinates of two contiguous residues are represented using polar coordinates. To encode $(\rho_i, \theta_i, \phi_i)$ into three distinct qubits, the coordinates must be normalized to the interval $[-1, 1]$ to ensure that the value of Pauli-Z (σ_z) expectation accurately represents the scaled polar coordinates. The range of ρ_i is defined by the maximum or minimum bond length

of the training dataset so that $\rho_i \in [v_{min} - \epsilon, v_{max} + \epsilon]$, where v represents the bond length of the training dataset and ϵ is a floating value set to $\epsilon = 0.02$. The intervals for θ and ϕ are $[0, \pi]$ and $[0, 2\pi]$, respectively, as defined. Subsequently, adjust the polar coordinates to incorporate the starting parameters $(\omega_\rho, \omega_\theta, \omega_\phi)$ as delineated in Equation 4.

$$\begin{cases} \omega_\rho = \arccos\left(2 \frac{\rho - v_{min} + \epsilon}{v_{max} - v_{min} + 2\epsilon} - 1\right) - \pi/2 \\ \omega_\theta = \arccos(2\theta/\pi - 1) - \pi/2 \\ \omega_\phi = \arccos(\phi/\pi - 1) - \pi/2 \end{cases} \quad (4)$$

Then, the basic circuit is depicted in Fig. 6. For an example of qubit represented by θ , the initial state is $|\psi\rangle = |0\rangle$. Following the Hadamard gate, the state is modified to Bell basis $|\psi\rangle = \frac{\sqrt{2}}{2} |0\rangle + \frac{\sqrt{2}}{2} |1\rangle$. Subsequently, $RY(\omega_\theta)$ transforms the state to Equation 5.

$$\begin{aligned} |\psi\rangle = & \frac{\sqrt{2}}{2} \frac{2\theta - \pi + 2\sqrt{-\theta(\theta - \pi)}}{\pi} |0\rangle \\ & + \frac{\sqrt{2}}{2} \frac{\pi - 2\theta + 2\sqrt{-\theta(\theta - \pi)}}{\pi} |1\rangle \end{aligned} \quad (5)$$

At this point, if the expected value of the qubit is computed with respect to σ_z , the outcome will be Equation 6.

$$E = \langle \psi | \sigma_z | \psi \rangle = \frac{2\theta - \pi}{\pi} \quad (6)$$

Following the measurement of the quantum circuit, the parameter θ will undergo post-processing defined by $\theta' = \frac{\pi}{2}(E + 1) = \theta$. The coordinates encoded by QCSE can be accurately reproduced without additional transformations, demonstrating that QCSE can fully encode coordinate information and possesses robust interpretability.

The QCSE encoder method implemented continuous coordinates rather than a discrete lattice encoder. Consequently, more precise structures will be predicted, and consistency will be ensured.

Model workflow of QSyncFold

The model is a classical-quantum hybrid that implements parameter processing and post-processing on a classical computer, encoding

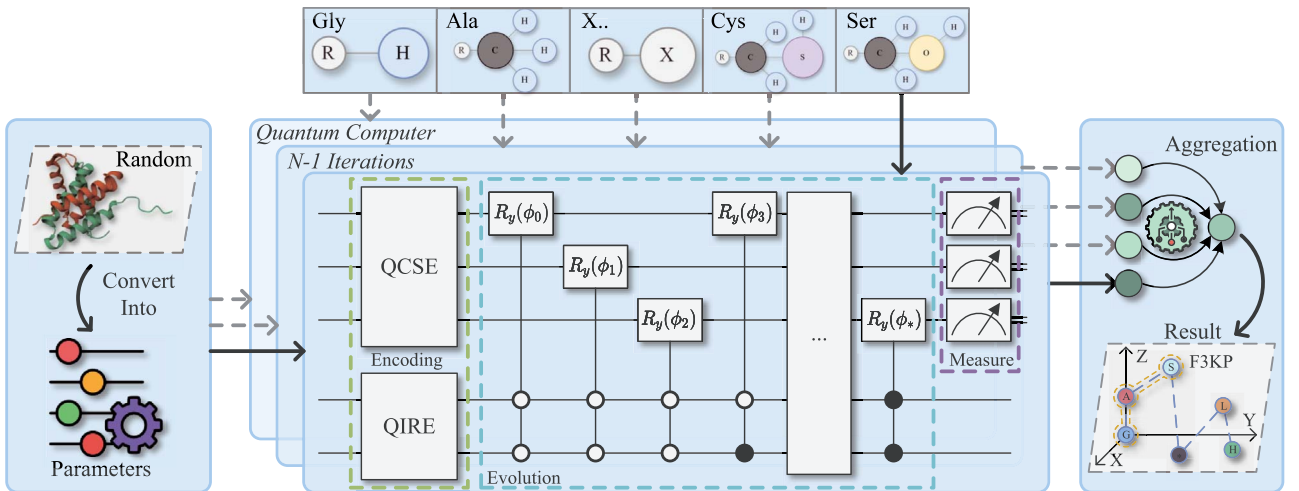


Figure 7. Topology of the QSyncFold model, which initiates with initial structures based on prior knowledge to compute circuit parameters, followed by the quantum circuit for each residue to optimize relative position, with classical structure then determined via quantum measurement and post-processing.

data and optimizing the residue structure on a quantum computer, and fine-tunes parameters using adjoint differentiation and gradient descent methods during training. The model primarily contains four components as shown in Fig. 7. The initial component is the random structure generation module, which employs the QCSE approach. In QCSE, prior knowledge of the training set is utilized to initialize the random structure effectively, detailed in [Supplementary E](#). By randomly generating dozens of structure batches for optimization, the jitter of the results can be eliminated to a certain extent. The second module calculates the parameters necessary for the quantum circuit. Alongside the pertinent values presented in QIRE and QCSE, other auxiliary parameters, including direction vectors and distance maps, are also computed. The subsequent component is the quantum circuit. Each residue r_i utilizes the relative position and interaction strength of other residues to optimize the positioning of residue r_i . The optimal protein structure can ultimately be predicted by $N - 1$ iterations for each residue (where the coordinates of residue r_0 are meaningless and fixed at $(0, 0, 0)$). The final stage is post-processing. Having been normalized before inputting, the coordinate data have to be restored during the measurement stage. Secondly, for some extreme value data, the first 25% and the last 25% of the dataset are eliminated, and the mean is computed to derive the ultimate relative orientation, which is subsequently transformed into Cartesian coordinates of the spatial structure.

The quantum circuit in QSyncFold primarily serves to optimize the position coordinates of residues. The circuit employs QIRE to encode interaction strength and QCSE for spatial coordinates. According to QCSE, three qubits are employed to encode the spatial polar coordinates (ρ, θ, ϕ) of the residue; that conversion is detailed in [Supplementary F](#). According to QIRE, $\lceil \log_2 N \rceil$ qubits are utilized to represent the interaction intensity among residues, with N being the maximal protein length supported by the model. After finishing the initialization of QIRE and QCSE, we proceed to the stage of structural adjustments. Using the adjustment of ρ for residue r_i as an illustration, the state corresponding to residue r_j is $\alpha_j |j\rangle$. Let the line associated with ρ be $|\psi\rangle$, and execute the transformation delineated in Equation 7.

$$\left\{ \begin{array}{l} |\psi\rangle_{next} = |\psi\rangle MCRY(\theta, CtrlQubits, CtrlValues) \\ \theta = Direct[i, j] \cdot k[j - i] \\ CtrlQubits = binary(j) \\ CtrlValues = \{\psi_i \mid i \in CtrlQubits\} \end{array} \right. \quad (7)$$

Execute the aforementioned calculations for each interaction and coordinate. Ultimately, the coordinate values modified by the model are obtained by assessing the expectation of the three qubits representing the coordinates on Pauli-Z. In training step, Adam optimizer [33] and the Adjoint quantum circuit differentiation method [34] are used. Due to the limitation of the NISQ era, both the quantum approaches focus on short peptides where multiple sequence alignments (MSAs) are scarce, and lead to the design without homologous sequences or co-evolutionary information. All experiments are conducted with the Pytorch deep learning framework [35] and the PennyLane quantum computer simulator [36].

In summary, the QSyncFold model accomplished the prediction of single-chain protein structures. This QSyncFold model, designed using QCSE and QIRE, comprehensively addresses the shortcomings of previous quantum models.

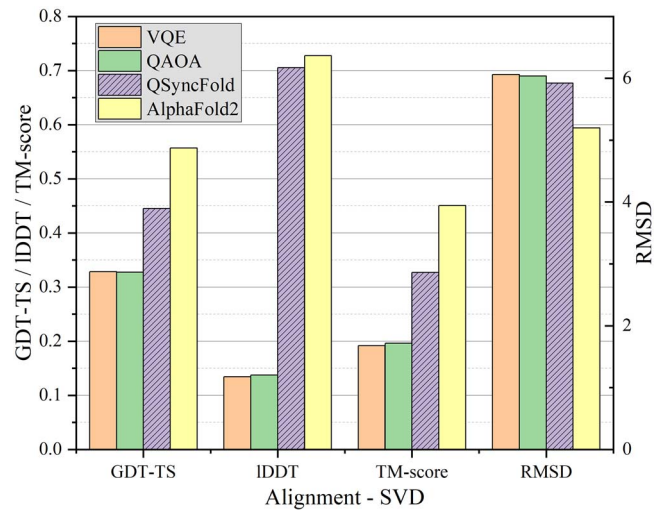


Figure 8. Performance comparison of QSyncFold and previous quantum models across all metrics.

Results

This experiment assesses the performance of four protein-folding models, including VQE [26], QAOA [37], AlphaFold2 [12], and QSyncFold, using four metrics: RMSD [38], GDT-TS [39], IDDT [40], and TM-score [41], across two alignment methods (SVD and F3KP in [Supplementary G](#) for detailed algorithms). Single-chain small proteins were filtered from the RCSB PDB database [42] to constitute the dataset. This dataset was partitioned into training and testing sets using an 8:2 ratio, with training validated through five-fold cross-validation. MSA-only searches with ColabFold did not identify additional homologous sequences for any test protein.

QSyncFold achieves superior performance to prior quantum models

Performance comparison. The results under SVD alignment are presented in Fig. 8, while the results under F3KP alignment are shown in [Supplementary H](#). Detailed data are provided in [Table 1](#). QSyncFold demonstrates superior performance among quantum models (VQE and QAOA). Across GDT-TS, IDDT, and TM-score, QSyncFold significantly outstrips VQE and QAOA, highlighting its effectiveness in leveraging quantum computing for PSP. For instance, QSyncFold achieves up to a 5.25-fold improvement over VQE in the IDDT metric using the F3KP alignment method. Compared with earlier quantum algorithms, the QNN-based QSyncFold exhibits substantial improvements, effectively harnessing quantum computing properties to excel in specific problem domains. The analysis results shows that AlphaFold2 only maintain a slight edge over QSyncFold even though it outperforms previous quantum models across all metrics. On all metrics, the gap between QSyncFold and AlphaFold2 is significantly smaller than that between VQE and AlphaFold2. Specifically, on the IDDT metric, QSyncFold differs from AlphaFold2 by merely 3.1%.

3D structure visualization. To illustrate differences in predicted protein structures, we analyze the protein with PDB ID 7n2i [43], a human single-chain fibrin comprising nine residues. As shown in the [Table 2](#), QSyncFold exhibits lower variance than AlphaFold2 for this protein, indicating greater prediction stability. Both the QSyncFold multifold model and the AlphaFold2 multi-model yield consistent

Table 1 Performance comparison

Model	Metric	RMSD	GDT-TS	IDDT	TM-score
QSyncFold	SVD	5.921	0.445	0.705	0.327
	F3KP	8.478	0.454	0.705	0.406
VQE	SVD	6.061	0.328	0.134	0.192
	Δ	2.362%↑	35.654%↑	425.195%↑	70.744%↑
	F3KP	10.018	0.280	0.134	0.223
	Δ	18.169%↑	62.162%↑	425.195%↑	81.875%↑
QAOA	SVD	6.036	0.327	0.137	0.196
	Δ	1.930%↑	35.977%↑	413.572%↑	66.795%↑
	F3KP	9.789	0.281	0.137	0.224
AlphaFold2	Δ	15.464%↑	61.532%↑	413.572%↑	81.309%↑
	SVD	5.197	0.557	0.728	0.451
	Δ	-12.232% ↓	-20.058% ↓	-3.093% ↓	-27.390% ↓
	F3KP	5.087	0.588	0.728	0.502
	Δ	-39.997% ↓	-22.884% ↓	-3.093% ↓	-19.212% ↓

Table 2 Performance comparison on protein of Pdb ID 7n2i

Model	Metric	RMSD	GDT-TS	IDDT	TM-score
VQE		13.944	0.222	0.089	0.170
QAOA		13.721	0.194	0.089	0.169
QSyncFold	kfold0	5.920	0.444	0.835	0.378
	kfold1	8.833	0.333	0.821	0.291
	kfold2	8.751	0.333	0.751	0.299
	kfold3	9.600	0.306	0.673	0.274
	kfold4	10.696	0.306	0.772	0.287
	mean	8.760	0.344	0.770	0.306
	variance	3.132✓	0.003✓	0.004	0.002✓
ColabFold	model0	3.640	0.639	0.904	0.492
	model1	8.519	0.417	0.935	0.358
	model2	5.936	0.472	0.903	0.415
	model3	4.793	0.528	0.906	0.428
	model4	5.562	0.500	0.951	0.416
	mean	5.690	0.511	0.920	0.422
	variance	3.272	0.007	0.000✓	0.002

results with minimal variation. The predicted 3D structures are compared with the natural crystal structure in Fig. 9. For VQE and QAOA, inter-residue distances are initially unit lengths, requiring scaling by the average bond length of 3.8 Å [44] for visualization clarity. In contrast, QSyncFold and AlphaFold2 predict structures directly at the correct scale. VQE predictions feature a loop with two overlapping residues, while QAOA produces a similar structure with increased overlap. These inaccuracies stem from lattice network encoding constraints, limiting growth directions and impairing prediction quality. Conversely, QSyncFold and AlphaFold2 accurately predict the protein's fundamental conformation across both average and sub-models. Both models excel at capturing fundamental secondary structures. Overall, QSyncFold outperforms other quantum models in aligning with the target structure, suggesting it better captures essential geometric features. QSyncFold effectively predicts

macroscopic protein conformations, exceeding the capabilities of existing quantum models, with minimal sub-model discrepancies.

ProtaQode ablation experiments. To validate the effectiveness of the quantum encoder ProtaQode, ablation experiments were conducted. Performance differences across various metrics were observed when removing the ProtaQode encoder or its sub-encoders QCSE or QIRE, as shown in Fig. 10 with SVD alignment and F3KP alignment in Supplementary H. The ablation results demonstrate that ProtaQode effectively enhances the capability of quantum encoding for proteins. The sub-encoder QCSE yields a substantial improvement in encoding performance, as QCSE employs an expectation-based encoding scheme to represent continuous coordinates, which provides stronger geometric expressiveness than previous limited discrete encoding approaches. Without QCSE, coordinate updates become constrained and geometric errors accumulate during iterative

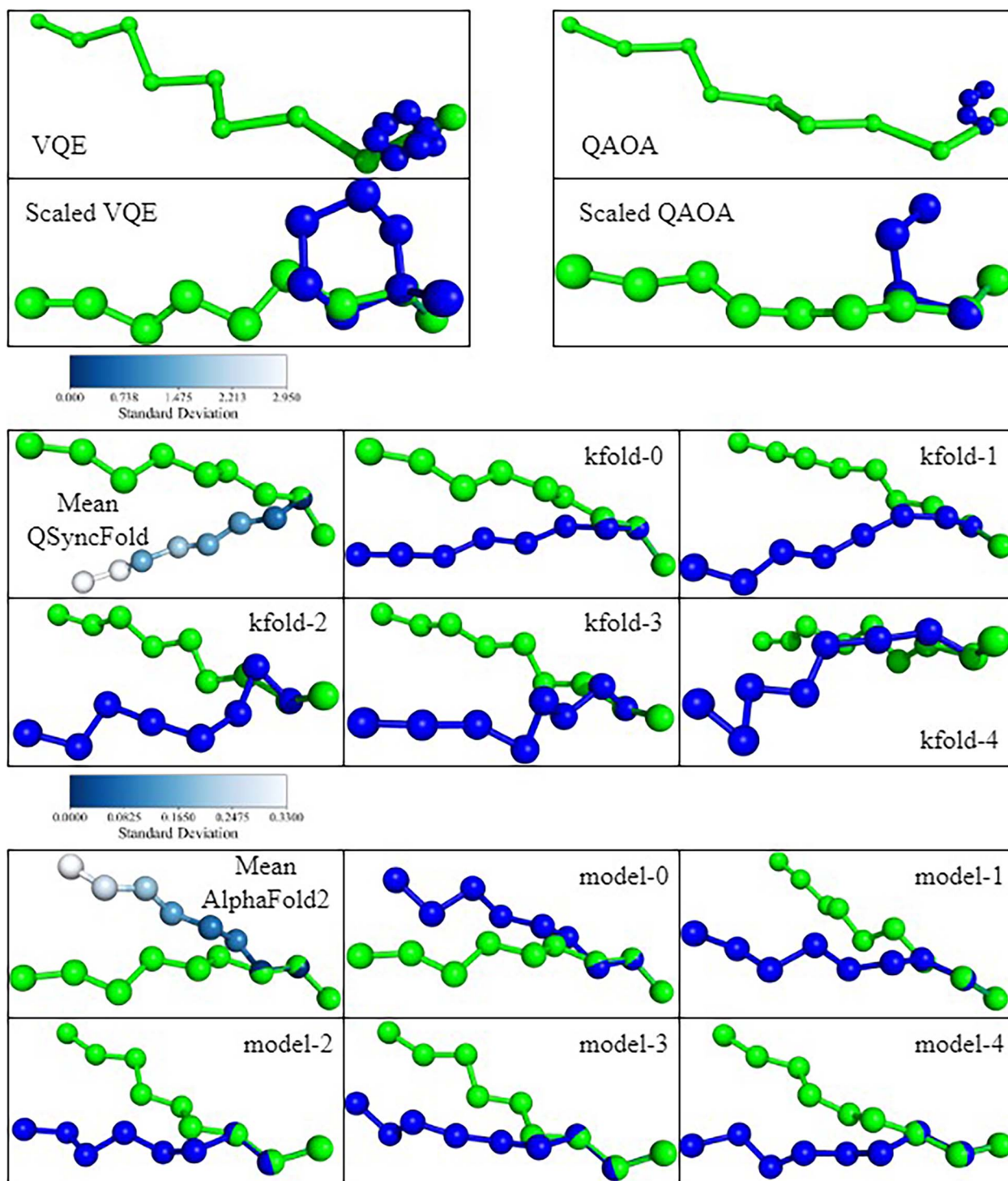


Figure 9. 3D structures of protein PDB ID 7n2i. The natural conformation of the protein is shown as a reference superimposed with the model-predicted conformations.

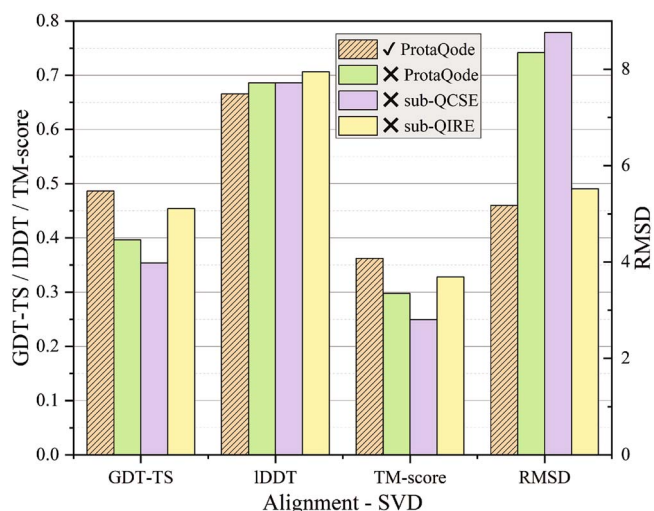


Figure 10. Ablation study of the ProtAQuery encoder.

optimization. In addition, the sub-encoder QIRE leverages quantum superposition and entanglement to learn and encode residue-residue interaction strengths, and its removal weakens the propagation of

long-range structural constraints, leading to consistent but less severe degradation across global structural metrics. In summary, comparative experiments, 3D case visualization, and ablation studies all validate the effectiveness of the QSyncFold model. QSyncFold accurately predicts protein structures, surpassing previous quantum models, thereby significantly advancing QML in PSP.

Scalability and resource analysis of QSyncFold

Stability with protein length. Figure 11 illustrates the trends of four performance metrics across four models as protein length increases. For each protein length, the shaded area spans the minimum to maximum metric values, highlighting the range of fluctuations. The QSyncFold model maintains high performance across all protein lengths. As protein length increases, the fluctuation range of QSyncFold decreases, resulting in a narrower shaded area. This demonstrates its superior prediction accuracy and robustness across varying protein lengths within the evaluated short-protein regime. AlphaFold2 exhibits similar performance, though QSyncFold shows slightly smaller variation. In contrast, the shaded areas for VQE and QAOA widen with increasing protein length, indicating reduced stability. Thus, QSyncFold outperforms VQE and QAOA in predictive stability and sustains high accuracy with longer protein sequences.

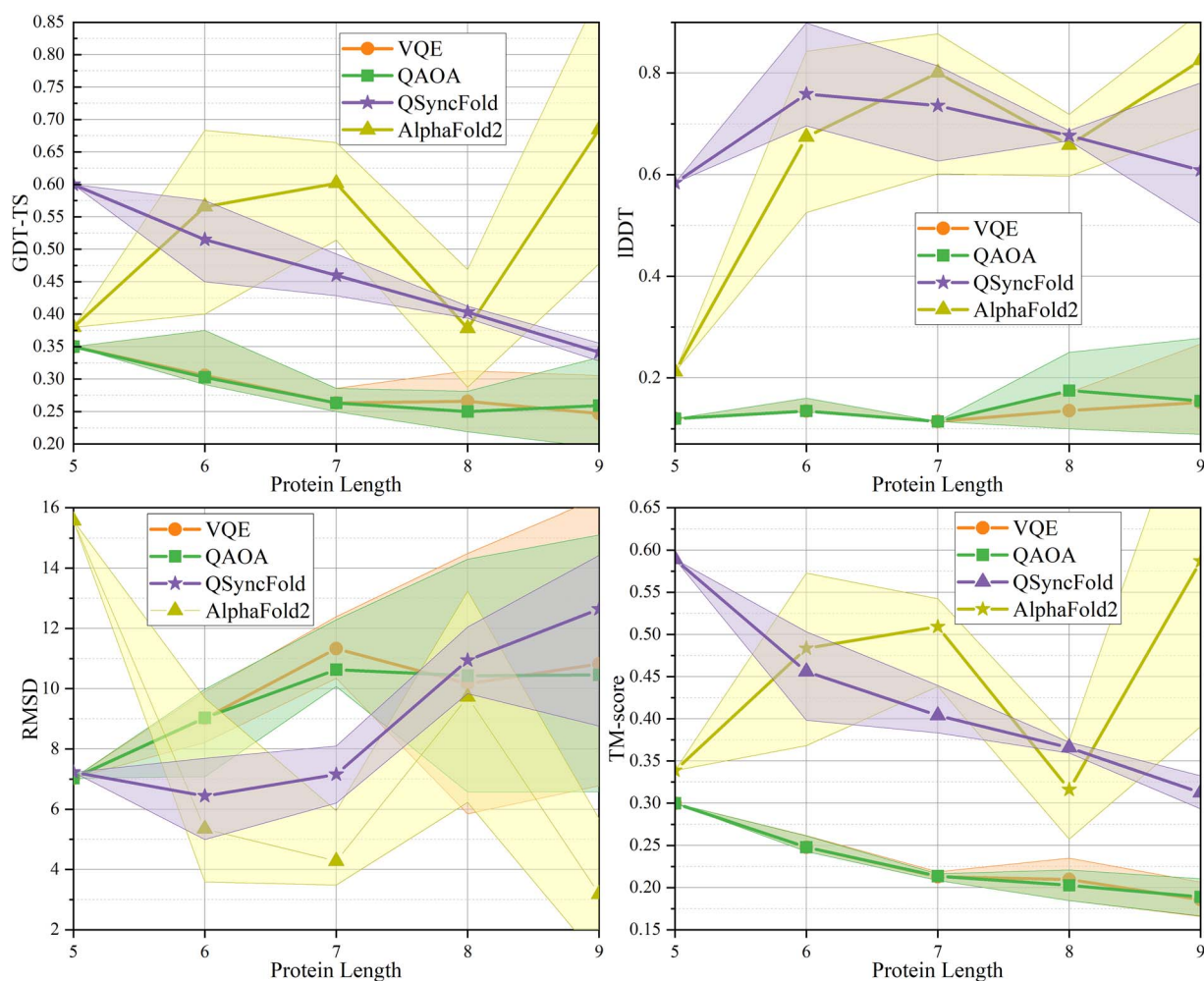


Figure 11. Performance and variation of each model across four metrics as protein length increases. The shaded area represents the variation in metrics.

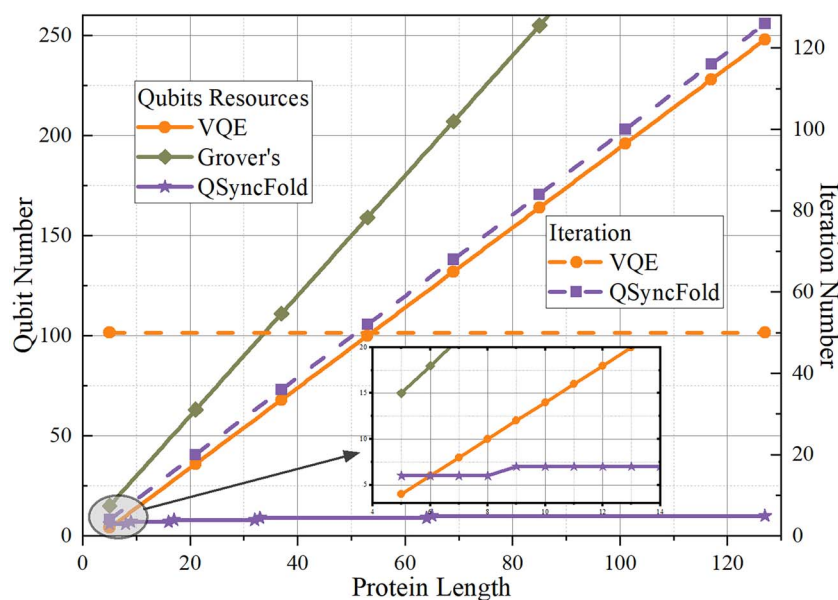


Figure 12. Comparison of resource consumption with other quantum approaches.

Resource consumption. A theoretical study on resource consumption was conducted to demonstrate the low resource requirements of QSyncFold compared with other QML models. When encoding coordinate information, the quantum coordinate state encoder (QCSE) uses a constant of 3 qubits. Additionally, for quantum interaction residue encoding (QIRE) with a maximum supported length of N , the use of quantum superposition reduces the required number of qubits to $\lceil \log_2 N \rceil$. As a result, the total number of qubits used in the quantum circuit per iteration is $3 + \lceil \log_2 N \rceil$. Figure 12 highlights the significant advantage of QSyncFold in quantum resource utilization, particularly in adapting to the limitations of the noisy intermediate-scale quantum (NISQ) era by conserving qubit resources. However, this efficiency comes at the cost of a linear increase in the number of iterations, which corresponds to the frequency of executing the quantum circuit.

Synchronization characteristics. QSyncFold fully leverages quantum properties to achieve synchronization characteristics. By entangling the interaction strength circuits in superposition states with the coordinate circuits, the model enables the coordinate circuits to explore multiple possible directions simultaneously at the same temporal instance, as illustrated in Fig. 13. The same exploration behavior is consistently observed across multiple proteins in the test set, as supported by the average-case directional analysis presented in Appendix K. This distinctive feature, made possible by the superposition principle of quantum computing, represents a significant advancement over classical methodologies.

Analysis of the biological significance of QSyncFold

QIRE encodes more interactions. Figure 14 demonstrates the superiority of our QIRE approach. The heatmap, with axes representing the protein sequence (PDB ID 7n2i), illustrates interaction strengths between residue pairs. Adjacent residues, primarily linked by peptide bonds, exhibit weak interactions with limited influence on global structure. In contrast, second-nearest neighbors display strong interactions, as corroborated by the Miyazawa–Jernigan matrix contact energy map [45]. Notably, QIRE also captures long-range interactions, which are essential for accurate PSP [46]. In summary,

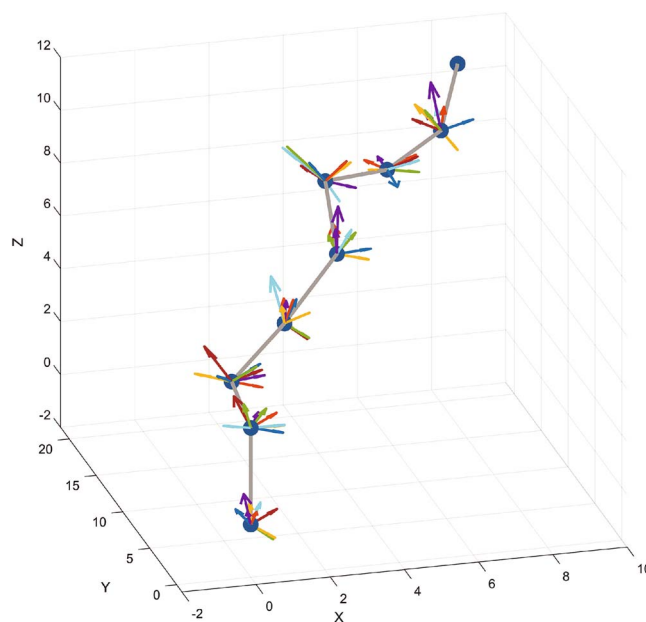


Figure 13. QSyncFold exhibits synchronization characteristics by superposition states.

QSyncFold effectively integrates local and long-range interaction data, with QIRE enhancing ability of the model to extract critical information for precise structural predictions.

Significance of interacting pairs. Although QSyncFold does not explicitly model biochemical properties, its quantum interaction states reveal biologically meaningful residue-level patterns. Quantum state analysis in Fig. 15 reveals the dominance of specific residue pairs across spatial scales. At hop=1, pairs V-K, A-K, and W-K exhibit the strongest interactions, whereas at hop=8, only W-K remains significant. This prevalence of short-range interactions highlights their critical role in protein folding dynamics. Lysine's positively charged ϵ -amino group facilitates electrostatic interactions, such as salt bridges and hydrogen bonds with acidic residues, stabilizing

Conclusion

We presented QSyncFold, a hybrid quantum–classical neural framework for PSP that explicitly targets three key limitations of prior quantum-PSP proposals: inefficient resource scaling, poor continuous-space encoding, and limited learnability. The core encoder, ProtaQode, couples the reversible QCSE for real-valued coordinate encoding with the QIRE that embeds adaptive interaction strengths in superposition via the ASRY gate. Algorithmically, QSyncFold realizes a practical resource–iteration trade-off: per-iteration qubit usage is reduced to $3 + \lceil \log_2 N \rceil$ for an N -residue chain.

Empirically, on our short-peptide dataset, QSyncFold substantially improves prediction quality relative to representative quantum baselines (VQE and QAOA), demonstrating concrete progress within the current landscape of quantum-based PSP methods. Depending on the alignment and evaluation settings, QSyncFold leads prior quantum methods across all metrics and achieves a 5.25-fold lead in the IDDT metric compared with the VQE method. Future work may explore integrating evolutionary information into quantum PSP, e.g. via quantum search algorithms such as Grover's method.

In summary, QSyncFold demonstrates that QNN can provide a resource-aware and interpretable path for quantum-native approaches to biomolecular modeling. While significant engineering and algorithmic work remains before wide applicability on realistic proteins is achieved, our results clarify the theoretical and practical trade-offs involved and point to concrete next steps for scaling quantum PSP as quantum hardware and algorithms continue to evolve.

Key Points

- QSyncFold introduces a hybrid quantum–classical neural network framework that integrates quantum superposition with differentiable learning for protein structure prediction.
- The proposed ProtaQode encoder achieves reversible continuous-space encoding via the Quantum Continuous Space Encoder (QCSE) and adaptive residue–pair interaction modeling via the Quantum Interactive Residue Encoder (QIRE).
- The model reduces qubit resource requirements from $O(N)$ to $3 + \lceil \log_2 N \rceil$ per iteration, making it feasible for current NISQ hardware.
- On short-peptide benchmarks, QSyncFold achieves up to a 5.25-fold improvement in IDDT over existing quantum baselines and narrows the performance gap to AlphaFold2.
- This work demonstrates a practical, resource-aware pathway for quantum-native protein folding and highlights the potential of quantum neural networks in biomolecular modeling.

Supplementary material

Supplementary material is available at *Briefings in Bioinformatics* online.

Conflicts of interest

None declared.

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Data availability

The data underlying this article are available in the article and in its online supplementary material. The code underlying this article are available in Zenodo, at <https://doi.org/10.5281/zenodo.18344057>. The datasets were derived from sources in the public domain: RCSB PDB, <https://www.rcsb.org/>.

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